

4. EDSAC (Electronic Delay Storage Automatic Calculator) was a pioneering British computer built at the University of Cambridge by Maurice Wilkes. Becoming operational in May 1949, it holds the distinction of being the first practical, full-scale, electronic stored-program computer to go into regular service.

Key Features:

- I. Memory: It used mercury delay lines to store data, holding 512 17-bit words (later upgraded to 1024 words).
- II. Speed: Capable of performing roughly 700 additions per second.
- III. Input/Output: Instructions were entered via 5-hole punched paper tape, and results were printed using a teleprinter

THE RISE OF THE DIGITAL AGE (From late 20th Century till date (21st Century))

Microprocessors (Early 1970s): The invention of the microprocessor (like the Intel 4004) put an entire CPU on a single chip. This led to personal computers.

Personal Computers (PCs) (Late 1970s - 1980s): Companies like Apple and IBM brought computing to homes and offices with machines like the Apple II and the IBM PC.

The Internet (Conceptualized 1960s, ARPANET 1969, World Wide Web 1989): The development of networking technologies and the creation of the World Wide Web by Tim Berners-Lee revolutionized communication and information access.

Graphical User Interfaces (GUIs) (1980s): Xerox PARC pioneered GUIs, which were popularized by Apple's Macintosh and later Microsoft Windows, making computers much more user-friendly.

Mobile Devices (1990s-2000s): Early mobile phones and PDAs evolved into smartphones, integrating computing power into devices we carry everywhere.

THE ERA OF BIG DATA, CLOUD, AND AI (21st Century)

Big Data: The explosion of digital information generated by internet use, sensors, and social media created the need to store, process, and analyze vast datasets.

Cloud Computing: Services like Amazon Web Services, Google Cloud, and Microsoft Azure provide scalable computing resources and storage over the internet, transforming how businesses and individuals use technology.

Machine Learning (ML) & Artificial Intelligence (AI): Driven by increased computing power, vast amounts of data, and sophisticated algorithms, ML and AI have seen remarkable progress. This includes advancements in:

Deep Learning: A subfield of ML inspired by the structure and function of the brain, leading to breakthroughs in image recognition, natural language processing, and more.

Natural Language Processing (NLP): Enabling computers to understand, interpret, and generate

human language (like this very conversation!).

Computer Vision: Allowing machines to "see" and interpret visual information from the world.